

Introduction to Programming

Quiz 1

Date: Thursday, 17 September 2026 Time: 17:00–17:45

Marks: 40 Weightage: 10%

Name: _____

Roll no.: _____

1. [10] Write a recursive function `dropOdds :: [a] -> [a]` that drops the elements at odd indices of a list (indexing from 0). For example:

`dropOdds [1,2,3,4,5] = [1,3,5]`

2. [10] Write a function `sums` that takes a list of lists of numbers `xs` and returns the list of sums of the non-empty lists in `xs`. For example:

`sums [[1..5], [], [2,6]] = [15,8]`

3. [10] Consider the function

`f xs = foldr (/=) False xs`

Find the type of `f`, and compute `f` on two example inputs.

4. [10] Given the decimal expansions of two numbers (as lists of digits of equal length, most significant digit first), compute the decimal expansion of their sum. For example:

`add [0,8,5,3,3] [0,2,8,6,8] = [1,1,4,0,1]`